Bruce Cheng

AI Application Engineer | Enterprise AI Architecture & LLM Systems
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Summary

Experienced in translating ambiguous enterprise requirements into production-ready AI systems and workflows, spanning on-prem LLM serving, RAG and agent workflows, multimodal document intelligence, and AI-enabled decision support. Strong in enterprise AI architecture, workflow orchestration, backend integration, and end-to-end solution delivery, with an emphasis on security, reliability, and practical adoption.

Experience

PixArt Imaging Inc. — Aug 2025 – Present
Information Engineer (Generative AI / LLM Projects) — *Hsinchu, Taiwan*

- Own end-to-end delivery of on-prem GenAI solutions for Legal, R&D, engineering, and internal knowledge scenarios in a semiconductor / IC design environment with strict data-security constraints.
- Designed and integrated a shared internal GenAI workspace around vLLM model serving, document/image utilities, and task-oriented interaction flows to support coding, document review, and technical analysis.
- Delivered secure analytics and document-intelligence systems spanning role-scoped NL2SQL, multimodal technical review, LLM operations reporting, and evidence-grounded RAG with page-level traceability.
- Productionized sensitive legal and patent workflows using OCR, local LLMs, staged restructuring/proofreading, terminology handling, and validation checks for reliable offline document processing.
- Automated engineering and meeting workflows with reusable Python-based tools, including robust file-processing pipelines, ASR, speaker-aware segmentation, and structured summarization.

AUO Corporation - Advanced Manufacturing Center — Aug 2021 – Aug 2025
AI Engineer / Project Leader — *Taichung, Taiwan*

- Led manufacturing-facing AI projects that connected models, data pipelines, explainability, and internal tools to real engineering and operational decision workflows.
- **Manufacturing Assistant Platform (RAG → Agent):** Evolved internal documentation retrieval into a domain-aware assistant capable of retrieval, SQL querying, API/tool use, structured diagnostics, and process-knowledge reasoning.
- **Cross-Factory Yield Optimization:** Developed a Golden Path optimization workflow and LIME-based explainable analysis to translate cross-factory variability into actionable engineering parameters, improving yield by 1.43%, reducing defects by 12.62%, and saving approx. 300 man-hours monthly.
- **Preventive Maintenance Workflow:** Designed an LSTM-based time-series prediction workflow with a custom trend-consistency objective, reducing defect prediction RMSE by 20% and saving approx. 90 man-hours monthly.
- Recognized as Smart Manufacturing Level 3 Elite Talent; preventive maintenance outcomes were approved for an internal defensive patent.

Academia Sinica - GIS Center — Feb 2019 – Jul 2019
Research Intern — *Taipei, Taiwan*

- Built a data quality workflow for parsing and normalizing ambiguous Taiwan address strings across heterogeneous research datasets.
- Developed rule-based normalization logic, structured address attributes, and a web-based sampling/review interface; won the Best Student Paper Award at TGIS 2019.

Selected Projects

Work-Hour Review & Insights Platform | *NL2SQL, Data Integration, Query Engine, FastAPI* — 2026

- Turned encrypted and legacy attendance data into manager-scoped dashboards and evidence-grounded NL2SQL analysis, using historical organization snapshots, server-enforced authorization, isolated read-only SQLite views, and audit logging.

Multimodal Technical Document Review | *VLM, MCP, Multimodal Pipeline, FastAPI* — 2026

- Built a format-preserving technical datasheet review workflow with multimodal analysis and Web, CLI, and MCP interfaces for scalable engineering review.

LLM Gateway & Operations Platform | *LLM Gateway, Model Routing, Usage Metering* 2025 – 2026

- Designed and built an internal LLM gateway and operations platform that centralized model access, API key management, routing, and usage metering, while consolidating multi-source telemetry into reporting for usage, cost, service operations, and executive decision support.

Evidence-Grounded Technical Knowledge | *VLM, RAG, Vector Search, Knowledge Extraction* 2026

- Created a human-reviewed knowledge pipeline that extracts engineering-document evidence and supports source-cited retrieval with exact page-level traceability.

Skills

Generative AI & LLM Systems: On-Prem LLM Serving, vLLM, RAG, Agent & Tool-Calling Workflows, Multimodal / VLM, OCR / ASR, Prompt & Validation Design
AI System & Solution Design: Enterprise AI Architecture, Workflow Orchestration, API / Tool Integration, Security & Reliability Design
Backend & Data Engineering: Python, FastAPI, REST APIs, SQL, Docker, ETL / Data Pipelines, Databricks
Applied Machine Learning: Time-Series Forecasting, LSTM, Explainable AI, Optimization

Education

National Yunlin University of Science and Technology 2019 – 2021
Master's Degree, Industrial Engineering and Management *Yunlin, Taiwan*

- **Thesis:** Improve the Predicted Efficiency of Crowd Flow in Large Areas Based on the R-tree and the SPACE-MDL-LSTM

National Yunlin University of Science and Technology 2015 – 2019
Bachelor's Degree, Industrial Engineering and Management *Yunlin, Taiwan*

- **Thesis:** Optimization of Machine Dispatching in Flexible Production Processes

Awards

Smart Manufacturing Level 3 Elite Talent | *AUO Corporation* 2025

Best Student Paper Award | *Taiwan Geographic Information Society Conference (TGIS)* 2020

Best Student Paper Award | *Taiwan Geographic Information Society Conference (TGIS)* 2019